

ch18 Refraction of Light

Items:

2018 Q17

2014 Q15

2020 Q13

2013 Q20

2021 Q13

2023 Q16

2016 Q20

2013 Q21

2025 Q14

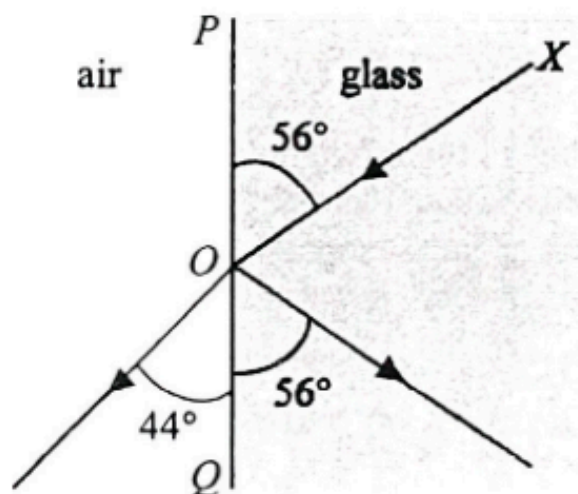
2016 Q17

2015 Q17

2024 Q14

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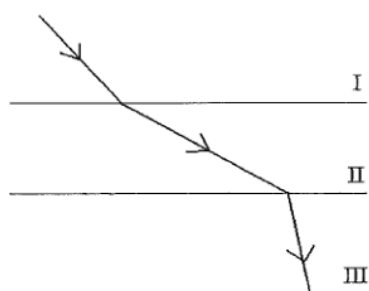
17.



In the figure above, XO is a light ray incident on the glass-air boundary. Which of the following gives the refractive index of glass?

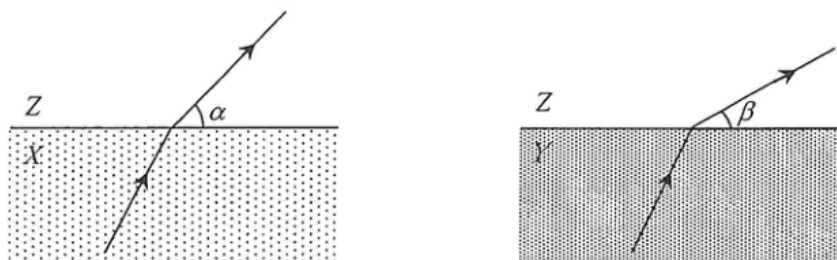
- A. $\frac{\sin 56^\circ}{\sin 44^\circ}$
- B. $\frac{\sin 44^\circ}{\sin 34^\circ}$
- C. $\frac{\sin 56^\circ}{\sin 46^\circ}$
- D. $\frac{\sin 46^\circ}{\sin 34^\circ}$

15. The figure shows the path of a light ray travelling from medium I to medium III separated by parallel boundaries. Arrange in **ascending order** the speed of light in the respective media.



- A. $I < III < II$
- B. $II < III < I$
- C. $III < I < II$
- D. $III < II < I$

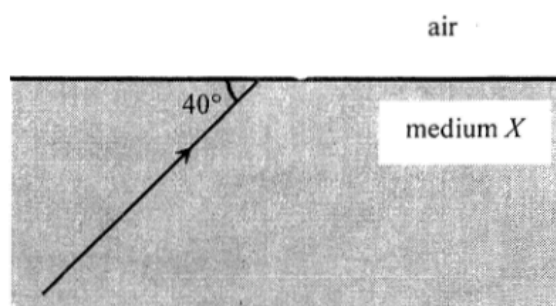
13. Monochromatic light travels with the same incident angle from media X and Y respectively to another medium Z as shown.



The corresponding refracted rays in Z make angles α and β respectively with the boundary plane (with $\alpha > \beta$). Which medium, X or Y , has a greater refractive index? In which medium, X or Y , does light travel faster?

	medium with a greater refractive index	medium in which light travels faster
A.	X	X
B.	X	Y
C.	Y	X
D.	Y	Y

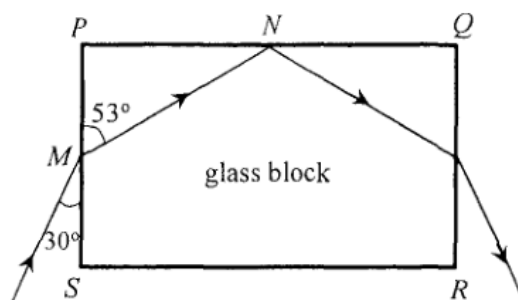
20.



A ray of light is travelling from a transparent medium X to air making an angle of 40° with the boundary plane as shown. If the angle between the refracted ray in air and the reflected ray in medium X is 70° , find the refractive index of medium X .

- A. $\frac{\sin 40^\circ}{\sin 30^\circ}$
B. $\frac{\sin 30^\circ}{\sin 40^\circ}$
C. $\frac{\sin 60^\circ}{\sin 50^\circ}$
D. $\frac{\sin 50^\circ}{\sin 60^\circ}$

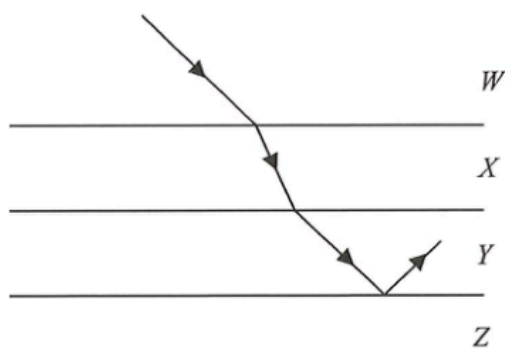
13.



The figure shows the cross-section of a rectangular glass block $PQRS$. A light ray is incident from air at M on face PS and the refracted ray strikes face PQ at N . Find the critical angle for the glass-air interface.

- A. 37°
- B. 44°
- C. 53°
- D. 60°

16. The figure shows a light ray passing through the horizontal boundaries between media W , X , Y and Z . The light ray in W is parallel to that entered into Y from X . Total internal reflection occurs at the boundary between Y and Z .



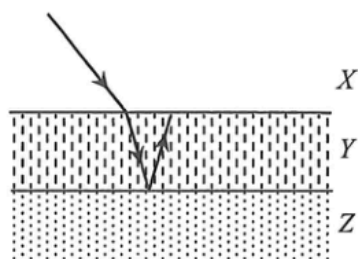
Which medium has the highest refractive index ?

- A. W
- B. X
- C. Y
- D. Z

20. A beam of white light is separated into different colours after entering a glass prism because lights of different colours
- A. are diffracted to different extents by the prism.
 - B. undergo total internal reflection at different angles inside the prism.
 - C. travel at different speeds in vacuum.
 - D. travel at different speeds in glass.

21. White light can be resolved into component colours by using a glass prism. Which of the following statements is/are correct ?
- (1) The refractive indices of glass for different component colours are not the same.
 - (2) Red light travels faster than violet light in a vacuum.
 - (3) The frequencies of all the component colours are reduced when entering the prism.
- A. (1) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (2) and (3) only

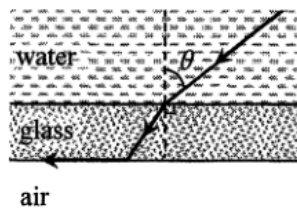
14.



The figure shows three media X , Y and Z separated by parallel boundaries. A light ray travelling from X into Y then undergoes total internal reflection at the boundary between Y and Z . Arrange in descending order the speed of light in the respective media.

- A. $X\ Y\ Z$
- B. $X\ Z\ Y$
- C. $Y\ Z\ X$
- D. $Z\ X\ Y$

17.

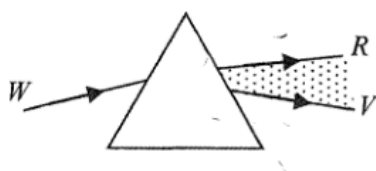


A parallel-sided glass sheet separates water from air. A ray of light in water is incident at an angle θ on the glass sheet and finally emerges into air along the glass-air interface as shown. Find θ .
Given: refractive index of water is 1.33.

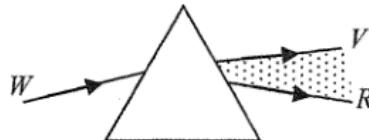
- A. 41.2°
- B. 48.8°
- C. 53.1°
- D. It depends on the refractive index of glass.

17. Which diagrams below correctly show the spectra formed from white light by a glass prism and a diffraction grating respectively? It is known that red light travels faster than violet light in glass.
(*R* = red, *V* = violet, *W* = white)

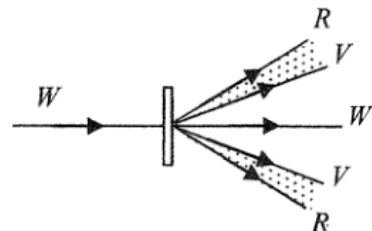
(1)



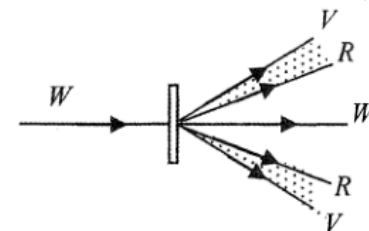
(2)



(3)

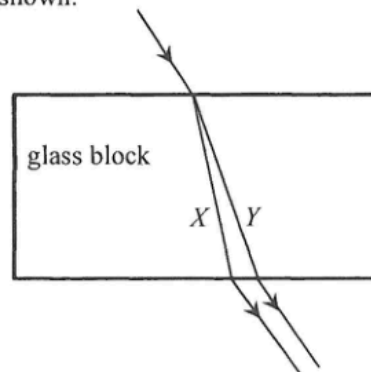


(4)



- A. (1) and (3) only
 B. (1) and (4) only
 C. (2) and (3) only
 D. (2) and (4) only

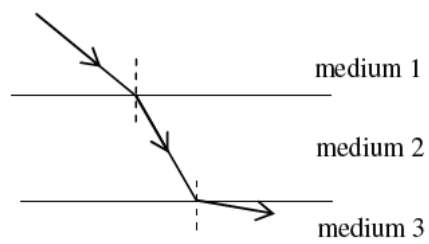
14. A light beam consisting of monochromatic light beam X and Y is incident on a rectangular glass block. It splits into two beams as shown.



Which of the following comparisons of beam X and Y is/are correct ?

- (1) X has a higher frequency.
 - (2) X has a larger speed in glass.
 - (3) In the glass block, the critical angle for X is greater.
- A. (1) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (2) and (3) only

20.



As shown above, a ray of light travels from medium 1 to medium 2, and then enters medium 3. The boundaries are parallel to each other. Arrange the speed of light, c , in the three media in **ascending** order.

- A. $c_3 < c_2 < c_1$
- B. $c_3 < c_1 < c_2$
- C. $c_2 < c_3 < c_1$
- D. $c_2 < c_1 < c_3$